README

Curve25519-family index calculus: first baseline (2026-11-02)

Class: first baseline for a new curve family (not an improvement claim, not a security claim about Curve25519). Protocol: PROTOCOL.md, frozen before data. Structure analysis: STRUCTURE.md. Report PDF: REPORT.pdf.

What was built

- `src/bin/curve25519_family.rs` (native Rust):
  - `audit`: re-derives Curve25519's facts at 255 bits from first principles (primality, `[ℓ](9) = O` by the x-only ladder, `#E = 8ℓ` by Hasse uniqueness, twist `4ℓ'` with a twist point, cyclic `Z/8` 2-Sylow, 4-torsion at `x = 1`, trace, `j`, embedding degree `> 10⁴`) → analysis/audit-255.json.
  - `family`: miniatures built by **Curve25519's own selection rule** (largest `p < 2^k`, `p ≡ 5 mod 8`; smallest `A ≡ 2 mod 4` with `#E = 8r`, `#E' = 4r'`, cyclic 2-part; base point the smallest `u` of order `r`), with BSGS point counting on curve and twist and the Montgomery summation polynomial `S₃` checked on 200 triples each → analysis/family.json.
  - `spec`: emits the frozen `ecbench.spec/v1` panels in `specs/`.
  - `summarise`: per-phase cost, trials per relation, prediction checks and the figure, from the session records.
  - Unit tests: BSGS order vs Legendre-sum count (curve and twist, every `A < 200` on three primes); `S₃` vanishes exactly on `x(P ± Q)`; the `T₀` symmetry `S₃(1/x₁, 1/x₂, x₃) = S₃(x₁, x₂, x₃)/(x₁x₂)²`.
- The index calculus itself is the lab's `ic.pipeline` (factor base → pair table oracle → relations → linear algebra → verified log), run unchanged on the miniatures in short Weierstrass form. Every solve is verified against the target.

k	p	A	#E = 8·r	#E' = 4·r'	base u
20	1048573	926	8·131203	4·261881	4
24	16777213	318	8·2097349	4·4193909	3
28	268435157	5330	8·33557809	4·67101961	23
32	4294967197	27842	8·536884639	4·1073714321	4

(At `k = 255` the same rule is the one that produced `A = 486662`, `u = 9`; the 255-bit search itself was not rerun, as it needs SEA point counting.)

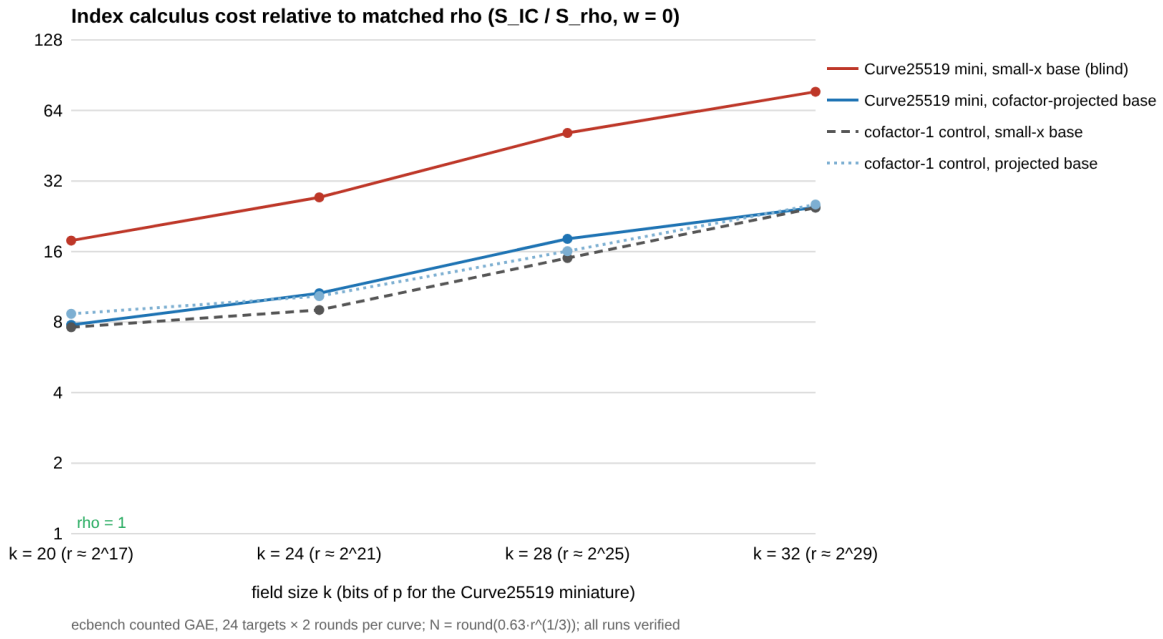
Result

ecbench, counted GAE, `S` = cost / `√r`; 24 public targets × 2 rounds per curve; 4,032 records, **all verified**; `ecbench verify --replay 12` audit OK on all four sessions (receipts below). Each session also measures a cofactor-1 control curve with a subgroup of the same size.

The Curve25519-family baseline (lowest `S` at `w = 0` among the four existing recipes, chosen in-sample as the protocol states) is `prime-projected-abscissa:size=N_a` + `mitm:negation_folded=1` at every size:

k	log₂ r	N	S baseline (w = 0 / 0.1 / 1)	S rho-neg (w = 0 / 0.1 / 1)	distance to rho (w = 0 / 0.1 / 1)
20	17.0	32	11.76 / 12.64 / 20.49	1.511 / 1.619 / 2.585	**7.8× / 7.8× / 7.9×**
24	21.0	81	12.57 / 13.69 / 23.79	1.184 / 1.290 / 2.251	**10.6× / 10.6× / 10.6×**
28	25.0	203	16.38 / 17.95 / 32.10	0.903 / 0.987 / 1.742	**18.1× / 18.2× / 18.4×**
32	29.0	512	23.31 / 25.52 / 45.41	0.944 / 1.037 / 1.871	**24.7× / 24.6× / 24.3×**

Rho is ahead at every size and weight (`ecbench band`: rho < every IC arm across the band). The gap grows with size, as expected of an `m = 2` generic decomposition whose cost scales as `r^{2/3}` against rho's `r^{1/2}`.



IC / rho by size

The Curve25519-specific effect: the cofactor costs blind IC a factor 8 in yield

k	trials / relation, small-x base	trials / relation, projected base	ratio (cofactor 8)	ratio (cofactor-1 control)	S small-x ÷ S projected (w = 0 / 0.1 / 1)
20	495	62	**7.95**	1.00	2.29 / 2.44 / 3.25
24	1362	163	**8.38**	1.00	2.57 / 2.74 / 3.57
28	3068	401	**7.66**	1.00	2.83 / 3.02 / 3.86
32	8224	1025	**8.02**	1.00	3.11 / 3.32 / 4.29

(`N = N_a`; at `N = 2N_a` the ratios are 7.88, 7.97, 7.94, 8.05.)

A small-abscissa base on a Curve25519-shaped curve is spread over all of `E(F_p) ≅ Z/8 × Z/r`; only one pair sum in eight lands in the subgroup holding the target, so the relation phase pays the cofactor in full while rho never does. Projecting the base by the cofactor removes exactly that factor (7.66–8.38 measured, 8 predicted) and leaves the control untouched (1.00). With the projection, the miniature behaves like a prime-order curve of the same `r` (IC ÷ rho 7.8 / 10.6 / 18.1 / 24.7 vs control 7.6 / 9.0 / 15.0 / 24.6).

Preregistered predictions

	prediction	outcome
P1	small-x ÷ projected trials ≈ 8 on the miniature (fail if < 4), ≈ 1 on the control (fail outside [0.67, 1.5])	**holds** at all 8 (k, N) cells
P2	miniature projected IC ÷ rho within 1.5× of control small-x IC ÷ rho	**holds**: 1.02, 1.18, 1.21, 1.00
P3	rho ahead at every size and weight	**holds**

Per-arm, per-curve detail with phase shares: `analysis/summary.md` (JSON); ecbench tables `analysis/table-k*.md`; band `analysis/band-k*.{txt,json}`.

Reading

- **What is measured:** the cost of the lab's best existing prime-field IC on curves with Curve25519's structure, and the size of the one structural effect that matters (cofactor 8). The baseline is now named for the ratchet.
- **What is not measured:** anything at 255 bits; any algebraic (Semaev system / PKM tower) decomposition; wall time (L0/L1/L2 mixed, local host, counts only).

- **Track:** generic decomposition (the pair-table oracle runs identically on a random base). The projected base is an existing lab method, so using it is the baseline, not an improvement.
- **The structure, in one line:** Curve25519's only IC-relevant special features are its cofactor 8 (handled: projection or fiber oracles) and the `x ↦ 1/x` symmetry of T_0 on the Kummer line (a factor `2^{m−1}` in `S_{m+1}` degree, unexploited so far). It has no small-norm endomorphism and a prime field with no subfield, so no known IC structure changes its exponent. See STRUCTURE.md.

Next improvements to try (ratchet candidates)

1. **Base size retune on the projected base.** The lab's prime-order optimum `N_a` was reused; `2N_a` is worse at every size, so the optimum may sit below `N_a`. A frozen holdout with a fresh `target_seed` decides it.
2. **Closed cofactor fiber for prime curves.** Port `ic_framework::fiber`'s `closed` mode (pair table holding `P_i + P_j + K`, `K ∈ E[8]`) to the prime pair-table oracle: at fixed unknowns it covers 8× the targets for 8× the table, which moves the balanced optimum by `8^{1/3} ≈ 2` in theory; it must beat `proj-a` on fresh targets across the band.
3. **T_0-closed base** (`{x} ∪ {1/x}` in Montgomery coordinates): the 2-element sub-case of 2, cheaper to build; measures whether the Kummer-line symmetry gives anything beyond projection.
4. **Algebraic track:** `pkm-semaev` on the miniatures, with S_3/S_4 symmetrised under the T_0 action.

Evidence

session	dir	records	audit receipt (`analysis/verify-k*.json`) SHA-256
ECBS1hd6199244a727	`sessions/k20`	1008	`e01a1a20bf443635a7b45ca0aef36906c5dd82f1447db503521c1bcd8bf43703`
ECBS1h6e348fb5e3ea	`sessions/k24`	1008	`bb04487d899d697781cebfcf3d252f350e9ddc560b5972c942c62a9f6de05722`
ECBS1hae69d3dfdd4d	`sessions/k28`	1008	`da7dbd3c9435d3eafb8d84cbd4a51fa1282663b23935d5e74e246908459bb3e3`
ECBS1h1c4978d3e652	`sessions/k32`	1008	`7818db19246f2d1ea69e90d28ce891620ae1e9c201d5796674b9db89ddf0ff90`

`records.jsonl` files live in the artifact store (`.artifacts.json` per session; `python3 tools/artifacts/artifacts.py pull research/curve25519_ic_baseline_20261102`).

Reproduce:

```
        cargo build --release --bin ecbench --bin curve25519_family
        for k in 20 24 28 32; do
          ./target/release/curve25519_family spec $k 24 $((2026110200+k)) | diff -
research/curve25519_ic_baseline_20261102/specs/c25519-k$k.json
          ./target/release/ecbench run --wait --spec
research/curve25519_ic_baseline_20261102/specs/c25519-k$k.json --out /tmp/k$k
        done
        ./target/release/curve25519_family summarise /tmp/summary /tmp/k20 /tmp/k24 /tmp/k28 /tmp/k32
```

Deviations and open items (stated, not hidden)

- **Route.** CLAUDE.md's default is Iso Lab on EKS via a director/executor task card. These sessions ran locally (one host, `ecbench host` in each `host.json`), without the card relay. Counted GAE is deterministic and host-independent and the replays reproduced, but no wall-time figure is claimed and **a second-host replay (isolab-eks) is required before any leaderboard or `benchmarks/` row** (validation gate). No such row is added here.
- **Curve registry.** The miniature and control slugs are not yet in `docs/curves/registry.json`; curves are identified by parameters above.
- **Conductor** was not installed in this container; no file reservation was taken.
- The baseline arm was chosen in-sample among four recipes; later candidates must be compared on fresh targets.

STRUCTURE

Curve25519: the structure an index calculus can see

Every fact marked **(verified)** is re-derived by `target/release/curve25519_family audit` (output `analysis/audit-255.json`) or by that binary's unit tests; facts marked *(literature)* are cited and were not re-derived here. Nothing in this note is a claim that Curve25519 is weaker than its rho bound.

1. The curve

$E : y^2 = x^3 + 486662 \cdot x^2 + x$ over F_p, $p = 2^{255} - 19$ (Bernstein, PKC 2006; RFC 7748). Birational to the twisted Edwards curve of Ed25519.

property	value	status
p prime, $p \bmod 8$	prime, 5	verified
#E	8ℓ, $\ell = 2^{252} + 27742317777372353535851937790883648493$ prime	verified: $[\ell](u=9) = 0$ by the x-only ladder, $[\ell-1](9) \neq 0$, and 8ℓ is the only multiple of ℓ in the Hasse interval
twist #E'	$4\ell'$, ℓ' prime (253 bits)	verified, incl. $[4\ell'](u=2) = 0$ on the twist
2-Sylow of E(F_p)	cyclic Z/8: $A^2 - 4$ is a non-square, so $T_0 = (0,0)$ is the only rational 2-torsion point	verified
rational 4-torsion	at $x = 1$ ($A + 2$ square, $A - 2$ non-square)	verified
trace $t = p + 1 - 8\ell$	$-2219385422189788282868155023270691879 62$ ($\lvert t \rvert \approx 2^{127.4}$)	verified
anomalous / supersingular	no / no	verified
embedding degree	$> 10^4$ (no $p^e \equiv 1 \bmod \ell$ for $e \leq 10^4$)	verified (bounded check)
j	not 0 or 1728: no GLV automorphism beyond ±1	verified
CM discriminant	squarefree part of $t^2 - 4p$ is large (SafeCurves "disc" criterion)	literature; needs a 257-bit factorisation, not re-derived
matched rho	$\approx 0.886 \cdot \sqrt{\ell} \approx 2^{125.8}$ group operations with the negation map	literature (SafeCurves)

2. What is special, and what each property does to index calculus

(a) Cofactor 8, cyclic. $E(F_p) \cong Z/8 \times Z/\ell$. The discrete log lives in the order-ℓ subgroup, so rho pays $\sqrt{\ell}$, but a factor base chosen by abscissa (Semaev's "small x", the lab's `prime-abscissa`) is spread over all 8ℓ points. In an $m = 2$ decomposition $R = \pm P_i \pm P_j$, only one pair sum in eight lands in the subgroup that holds R, so the per-probe relation yield is $\approx 2N^2/p$ rather than $2N^2/\ell$: the cofactor is paid in full by the relation phase and not at all by rho. Two existing remedies: `prime-projected-abscissa` (multiply each base point by 8 first) and the cofactor-fiber oracles of `ic_framework::fiber` (probe all eight lifts $R + K$, $K \in E[8]$), the latter so far implemented for binary curves only. This is the effect measured by prediction P1 of the protocol.

(b) The Kummer line and T_0. Montgomery's x-only arithmetic works on $E/\pm 1 \cong P^1$. Translation by T_0 acts there as $x \mapsto 1/x$ **(verified)**, and since $(P_1 + T_0) + (P_2 + T_0) = P_1 + P_2$, $S_3(1/x_1, 1/x_2, x_3) = S_3(x_1, x_2, x_3)/(x_1x_2)^2$ **(verified)**, where the Montgomery summation polynomial is

$$S_3(x_1, x_2, x_3) = (x_1 - x_2)^2 \cdot x_3^2 - 2[(x_1x_2 + 1)(x_1 + x_2) + 2A \cdot x_1x_2] \cdot x_3 + (x_1x_2 - 1)^2$$

(verified to vanish exactly on $x(P_1 \pm P_2)$). This is the prime-field, Montgomery-model instance of the 2-torsion symmetry that Faugère, Gaudry, Huot and Renault (J. Cryptology 2014) used to cut the degree of S_{m+1} for twisted Edwards / Jacobi models over F_{q^n}. Only $E[2]$ acts on the Kummer line (the order-4 and order-8

points do not commute with ±1), so the x-only symmetry group is $Z/2$ per pair of summands: a factor 2^{m-1} in the degree of S_{m+1} after symmetrisation, and a factor 2 in factor-base coverage at fixed unknowns when a base is closed under $x \mapsto 1/x$.

(c) No useful endomorphisms. $j \neq 0, 1728$ and a large CM discriminant: the only cheap automorphism is negation, which both rho (√2) and the IC pair table (`negation_folded=1`) already use. There is no GLV orbit to fold the factor base by, unlike the $j = 0$ curves in `research/generator_multiplier_e2e_20261101`.

(d) Prime field, no subfield. F_p has no proper subfield and no F_p-subspace structure to Weil-descend over, so the Gaudry–Diem route that gives subexponential IC over F_{q^n} does not exist. Decomposition over F_p reduces to finding small solutions of $S_{m+1}(x_1, \ldots, x_m, x_R) = 0$ (Semaev 2004); Petit–Kosters–Messeng (PKC 2016) replace "small" by a tower of rational-map images, which the lab has as `pkm-semaev`. No sub-$\sqrt{\ell}$ algorithm is known for any prime-field curve; the lab's prime-field ladder has rho ahead at every size (`docs/ic/LEADERBOARD.md`, Table B).

(e) $p \equiv 5 \pmod 8$. -1 is a square, 2 is not; square roots take one exponentiation (Atkin), which is the root-finding cost per trial of an $m = 2$ S_3 decomposition. It does not change asymptotics.

(f) Twist security and x-only. The twist's $4\ell'$ protects the ladder from invalid-point attacks. It does not touch the DLP on E, but it means a "curve or twist" factor base (both halves of the Kummer line) has no cheap second DLP to exploit.

3. The open-literature situation

There is no index-calculus paper aimed at Curve25519 specifically; the relevant work is general: Semaev (2004) summation polynomials; Gaudry (2009) and Diem (2011) over extension fields; Faugère–Gaudry–Huot–Renault (2014) torsion symmetries; Petit–Kosters–Messeng (2016) prime-field towers; Galbraith–Gaudry (2016) survey; Bernstein–Lange–Schwabe (2011) on the negation map in rho. The two properties of Curve25519 that interact with IC at all are (a) and (b); neither changes the exponent of a prime-field IC.

PROTOCOL

Curve25519-family index-calculus baseline — protocol

Frozen 2026-11-02 before any ecbench data was taken. Amendments, if any, are appended below with their reason and never rewrite this text.

Question

What does the lab's existing prime-field index calculus cost on curves that carry Curve25519's structure, and how far is it from matched Pollard rho? This is the **first baseline** for the Curve25519 family (AGENTS.md "Success criterion": name the baseline before optimising). It is not an improvement claim and not a security claim about Curve25519.

Curves

Curve25519 is $y^2 = x^3 + 486662x^2 + x$ over F_p, $p = 2^{255} - 19$. Its facts are re-derived from first principles by `cargo run --release --bin curve25519_family -- audit` (output: `analysis/audit-255.json`). No index calculus runs at 255 bits, so the measured curves are **miniatures built by Curve25519's own selection rule** (`curve25519_family family`, `analysis/family.json`):

1. p = the largest prime below 2^k with $p \equiv 5 \pmod 8$ (as $2^{255} - 19$).
2. Montgomery form $B = 1$, $A \equiv 2 \pmod 4$, $A \geq 6$ (integral $a24 = (A-2)/4$).
3. $A^2 - 4$ a non-square, so $T_0 = (0,0)$ is the only rational 2-torsion point and the 2-Sylow subgroup is cyclic $Z/8$, as for Curve25519.
4. $\#E = 8r$, $\#E' = 4r'$ (twist), r, r' prime, $r > 2^{k-3}$.
5. A the smallest value meeting 1–4; base point the smallest $u \geq 2$ of order r (Curve25519: $u = 9$).

Point counts are by baby-step giant-step on curve and twist, unit-tested against the Legendre-sum count; the Montgomery summation polynomial S_3 is checked on 200 random triples per curve and unit-tested to vanish exactly on $x(P \pm Q)$.

Sizes: $k \in \{20, 24, 28, 32\}$ (r of 18, 22, 26, 30 bits). Each session also carries a **cofactor-1 control**: `prime_search` at the bit length of that miniature's r, so the rho reference costs the same on both curves and any difference in IC ÷ rho isolates the cofactor (and Montgomery-model) structure.

Arms (identical on both curves of a session)

arm	role	method
`rho-neg`	reference	`rho.negation`
`abs-a`, `abs-2a`	baseline	`ic.pipeline`, `prime-abscissa:size=N`, `mitm:negation_folded=1`
`proj-a`, `proj-2a`	baseline	`ic.pipeline`, `prime-projected-abscissa:size=N,scan_cap=16N`, `mitm:negation_folded=1`
`abs-2a-aa`	control (A/A)	copy of `abs-2a`
`rho-neg-aa`	control (A/A)	copy of `rho-neg`

$N_a = \text{round}(0.63 \cdot r^{1/3})$, the lab's validated prime-order optimum (`docs/ic/LEADERBOARD.md`, "Prime-field baseline": 162, 407, 1024 at 24, 28, 32 bits) scaled to the subgroup. `mitm:negation_folded=1` is the incumbent

exact oracle that holdout selected at every size. All arms are existing lab methods; nothing new is introduced in the pipeline.

Targets: 24 public targets per curve, 2 measured rounds, 1 warm-up, alternating arm order, seeds in the specs. Specs are emitted by `curve25519_family spec <k> 24 <seed>`, seed `2026110200 + k`.

Unit, isolation, pricing

`S` = counted cost in GAE per `√r` (ecbench). Isolation **L0**, counted units only: no wall-clock figure is claimed (AGENTS.md §12: levels gate wall time, never counts). Every IC-vs-rho ratio is reported at the §5a band `w = 0, 0.1, 1` GAE per unpriced native operation (`ecbench band`); an order between two arms is stated only where it holds across the band.

Predictions (preregistered)

Mechanism. `prime-abscissa` points lie anywhere in `E(F_p) ≅ Z/8 × Z/r`, so only 1/8 of their pair sums land in the order-`r` subgroup that holds the targets: per-probe yield ≈ $2N^2/(8r) = 2N^2/p$. `prime-projected-abscissa` multiplies each base point by the cofactor first, so every pair sum is a subgroup point: yield ≈ $2N^2/r$.

- **P1** (cofactor penalty). On the miniature, `abs-*` needs ≈ 8× the relation-phase trials of `proj-*` at equal `N`; on the cofactor-1 control the ratio is ≈ 1 (projection by `h = 1` is the identity).
- **P2** (control matches). Miniature `proj-*` IC ÷ rho is within a factor 1.5 of control `abs-*` at the same `N`.
- **P3** (distance). Rho stays ahead at every size and weight (the lab's prime-order ratio is 7.7–33×); the ratio is reported as distance remaining.

Falsification: P1 fails if the miniature `abs/proj` trial ratio is below 4 or the control's is outside `[0.67, 1.5]`.

Baseline definition (the ratchet's first entry)

Per size, the **Curve25519-family baseline** is the arm with the lowest mean `S` at `w = 0` among `abs-a, abs-2a, proj-a, proj-2a` on the miniature, with its `w = 0.1, 1` values. This is an in-sample choice among four existing recipes and is disclosed as such; any later candidate must beat it on fresh targets (a new `target_seed`) in a new frozen protocol.

Acceptance

Every record verified; `ecbench verify --replay 12` audit passes on every session; A/A ratio within its interval. A failed or timed-out arm is reported, not dropped.

Execution route and its limits

CLAUDE.md names Iso Lab on EKS as the default route and an S3 resume checkpoint before runs. The resume checkpoint was pushed (`s3://setec-astronomy-artifacts/checkpoints/crypto/claude__curve25519-index-calculus-pw5d6r/20261102T010052Z-1b3c6210ac90/`). This container has no `aws` CLI and no Iso Lab identity, so the sessions run locally at L0 with counted units only (the same route as `research/ic_residual_fused_20261031`, Amendment 1). A second-host replay on isolab-eks is an open item before any leaderboard row.